

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12406717
Kind Code	B2
Date of Patent	September 02, 2025
Inventor(s)	Ayyapureddi; Sujeet et al.

Apparatuses and methods for dynamically allocated aggressor detection

Abstract

Apparatuses, systems, and methods for dynamically allocated aggressor detection. A memory may include an aggressor address storage structure which tracks access patterns to row addresses and their associated bank addresses. These may be used to determine if a row and bank address received as part of an access operation are an aggressor row and bank address. The aggressor row address may be used to generate a refresh address for a bank identified by the aggressor bank address. Since the aggressor storage structure tracks both row and bank addresses, its storage space may be dynamically allocated between banks based on access patterns to those banks.

Inventors:	Ayyapureddi; Sujeet (Boise, ID), Morgan; Donald M. (Meridian, ID)
Applicant:	MICRON TECHNOLOGY, INC. (Boise, ID)
Family ID:	82405244
Assignee:	MICRON TECHNOLOGY, INC. (Boise, ID)
Appl. No.:	17/932206
Filed:	September 14, 2022

Prior Publication Data

Document Identifier	Publication Date
US 20230010619 A1	Jan. 12, 2023

Related U.S. Application Data

continuation parent-doc US 17153555 20210120 US 11482275 child-doc US 17932206

Publication Classification

Int. Cl.: G11C11/4078 (20060101); **G11C11/406** (20060101); **G11C11/408** (20060101)

U.S. Cl.:

CPC **G11C11/4078** (20130101); **G11C11/406** (20130101); **G11C11/4085** (20130101);
 G11C11/4087 (20130101);

Field of Classification Search

CPC: G11C (11/4078); G11C (11/406); G11C (11/4085); G11C (11/4087)

USPC: 365/149

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
3633175	12/1971	Harper	N/A	N/A
4679173	12/1986	Sato	N/A	N/A
5089957	12/1991	Stultz et al.	N/A	N/A
5291198	12/1993	Dingwall et al.	N/A	N/A
5299159	12/1993	Balistreri et al.	N/A	N/A
5422850	12/1994	Sukegawa et al.	N/A	N/A
5617348	12/1996	Maguire	N/A	N/A
5638317	12/1996	Tran	N/A	N/A
5699297	12/1996	Yamazaki et al.	N/A	N/A
5751655	12/1997	Yamazaki et al.	N/A	N/A
5768196	12/1997	Bloker et al.	N/A	N/A
5874907	12/1998	Craft	N/A	N/A
5933377	12/1998	Hidaka	N/A	N/A
5943283	12/1998	Wong et al.	N/A	N/A
5970507	12/1998	Kato et al.	N/A	N/A
5999471	12/1998	Choi	N/A	N/A
6002629	12/1998	Kim et al.	N/A	N/A
6011734	12/1999	Pappert	N/A	N/A
6061290	12/1999	Shirley	N/A	N/A
6212118	12/2000	Fujita	N/A	N/A
6310806	12/2000	Higashi et al.	N/A	N/A
6317381	12/2000	Gans et al.	N/A	N/A
6373738	12/2001	Towler et al.	N/A	N/A
6392952	12/2001	Chen et al.	N/A	N/A
6424582	12/2001	Ooishi	N/A	N/A
6434064	12/2001	Nagal	N/A	N/A
6452868	12/2001	Fister	N/A	N/A
6480931	12/2001	Buti et al.	N/A	N/A
6515928	12/2002	Sato et al.	N/A	N/A
6567340	12/2002	Nataraj et al.	N/A	N/A
6950364	12/2004	Kim	N/A	N/A
7027343	12/2005	Sinha et al.	N/A	N/A
7057960	12/2005	Fiscus et al.	N/A	N/A

7082070	12/2005	Hong	N/A	N/A
7187607	12/2006	Koshikawa et al.	N/A	N/A
7203113	12/2006	Takahashi et al.	N/A	N/A
7203115	12/2006	Eto et al.	N/A	N/A
7209402	12/2006	Shinozaki et al.	N/A	N/A
7215588	12/2006	Lee	N/A	N/A
7283380	12/2006	Srinivasan et al.	N/A	N/A
7304875	12/2006	Lien et al.	N/A	N/A
7319602	12/2007	Srinivasan et al.	N/A	N/A
7444577	12/2007	Best et al.	N/A	N/A
7551502	12/2008	Dono et al.	N/A	N/A
7565479	12/2008	Best et al.	N/A	N/A
7830742	12/2009	Han	N/A	N/A
7870362	12/2010	Hong et al.	N/A	N/A
7872907	12/2010	Okayama et al.	N/A	N/A
8174921	12/2011	Kim et al.	N/A	N/A
8400805	12/2012	Yoko	N/A	N/A
8451677	12/2012	Okahiro et al.	N/A	N/A
8625360	12/2013	Iwamoto et al.	N/A	N/A
8676725	12/2013	Lin et al.	N/A	N/A
8681578	12/2013	Narui	N/A	N/A
8756368	12/2013	Best et al.	N/A	N/A
8811100	12/2013	Ku	N/A	N/A
8862973	12/2013	Zimmerman et al.	N/A	N/A
8938573	12/2014	Greenfield et al.	N/A	N/A
9032141	12/2014	Bains et al.	N/A	N/A
9047978	12/2014	Bell et al.	N/A	N/A
9058900	12/2014	Kang	N/A	N/A
9087554	12/2014	Park	N/A	N/A
9087602	12/2014	Youn et al.	N/A	N/A
9117544	12/2014	Bains et al.	N/A	N/A
9123447	12/2014	Lee et al.	N/A	N/A
9153294	12/2014	Kang	N/A	N/A
9190137	12/2014	Kim et al.	N/A	N/A
9190139	12/2014	Jung et al.	N/A	N/A
9251885	12/2015	Greenfield et al.	N/A	N/A
9286964	12/2015	Halbert et al.	N/A	N/A
9299457	12/2015	Chun et al.	N/A	N/A
9311985	12/2015	Lee et al.	N/A	N/A
9324398	12/2015	Jones et al.	N/A	N/A
9384821	12/2015	Bains et al.	N/A	N/A
9390782	12/2015	Best et al.	N/A	N/A
9412432	12/2015	Narui et al.	N/A	N/A
9418741	12/2015	Ramaraju et al.	N/A	N/A
9424907	12/2015	Fujishiro	N/A	N/A
9449675	12/2015	Wheeler	N/A	N/A
9484079	12/2015	Lee	N/A	N/A
9514850	12/2015	Kim	N/A	N/A
9570143	12/2016	Lim et al.	N/A	N/A
9646672	12/2016	Kim et al.	N/A	N/A

9672889	12/2016	Lee et al.	N/A	N/A
9685240	12/2016	Park	N/A	N/A
9691466	12/2016	Kim	N/A	N/A
9697913	12/2016	Mariani et al.	N/A	N/A
9734887	12/2016	Tavva	N/A	N/A
9741409	12/2016	Jones et al.	N/A	N/A
9741447	12/2016	Akamatsu	N/A	N/A
9747971	12/2016	Bains et al.	N/A	N/A
9761297	12/2016	Tomishima	N/A	N/A
9786351	12/2016	Lee et al.	N/A	N/A
9799391	12/2016	Wei	N/A	N/A
9805782	12/2016	Liou	N/A	N/A
9805783	12/2016	Ito et al.	N/A	N/A
9818469	12/2016	Kim et al.	N/A	N/A
9847118	12/2016	Won	N/A	N/A
9865326	12/2017	Bains et al.	N/A	N/A
9865328	12/2017	Desimone et al.	N/A	N/A
9922694	12/2017	Akamatsu	N/A	N/A
9934143	12/2017	Bains et al.	N/A	N/A
9953696	12/2017	Kim	N/A	N/A
10032501	12/2017	Ito et al.	N/A	N/A
10083737	12/2017	Bains et al.	N/A	N/A
10090038	12/2017	Shin	N/A	N/A
10134461	12/2017	Bell et al.	N/A	N/A
10147472	12/2017	Jones et al.	N/A	N/A
10153031	12/2017	Akamatsu	N/A	N/A
10170174	12/2018	Ito et al.	N/A	N/A
10176860	12/2018	Mylavarapu	N/A	N/A
10210925	12/2018	Bains et al.	N/A	N/A
10297305	12/2018	Moon et al.	N/A	N/A
10339994	12/2018	Ito et al.	N/A	N/A
10381327	12/2018	Ramachandra et al.	N/A	N/A
10387276	12/2018	Ryu et al.	N/A	N/A
10446216	12/2018	Oh et al.	N/A	N/A
10490251	12/2018	Wolff	N/A	N/A
10600462	12/2019	Augustine et al.	N/A	N/A
10600491	12/2019	Chou et al.	N/A	N/A
10607686	12/2019	Akamatsu	N/A	N/A
10629286	12/2019	Lee et al.	N/A	N/A
10679710	12/2019	Hirashima et al.	N/A	N/A
10705900	12/2019	Jin	N/A	N/A
10770127	12/2019	Shore et al.	N/A	N/A
10811066	12/2019	Jones et al.	N/A	N/A
10832792	12/2019	Penney et al.	N/A	N/A
10861519	12/2019	Jones et al.	N/A	N/A
10867660	12/2019	Akamatsu	N/A	N/A
10930335	12/2020	Bell et al.	N/A	N/A
10943636	12/2020	Wu et al.	N/A	N/A
10950289	12/2020	Ito et al.	N/A	N/A
10964378	12/2020	Ayyapureddi et al.	N/A	N/A

11011215	12/2020	Parry et al.	N/A	N/A
11043254	12/2020	Enomoto et al.	N/A	N/A
11139015	12/2020	Brown et al.	N/A	N/A
11152050	12/2020	Morohashi et al.	N/A	N/A
11158364	12/2020	Penney et al.	N/A	N/A
11158373	12/2020	Penney et al.	N/A	N/A
11200942	12/2020	Jenkinson et al.	N/A	N/A
11222682	12/2021	Enomoto et al.	N/A	N/A
11257535	12/2021	Shore et al.	N/A	N/A
11264096	12/2021	Schreck et al.	N/A	N/A
11322192	12/2021	Morohashi et al.	N/A	N/A
11361808	12/2021	Bell et al.	N/A	N/A
11386946	12/2021	Ayyapureddi et al.	N/A	N/A
11398265	12/2021	Wu et al.	N/A	N/A
11424005	12/2021	Penney et al.	N/A	N/A
11462291	12/2021	Pan	N/A	N/A
11482275	12/2021	Ayyapureddi et al.	N/A	N/A
11521669	12/2021	Enomoto et al.	N/A	N/A
11568918	12/2022	Ayyapureddi et al.	N/A	N/A
11600314	12/2022	Ayyapureddi et al.	N/A	N/A
11600326	12/2022	Schreck et al.	N/A	N/A
11664063	12/2022	Lovett	N/A	N/A
11688451	12/2022	Zhang et al.	N/A	N/A
11694738	12/2022	Morohashi et al.	N/A	N/A
11699476	12/2022	Brown et al.	N/A	N/A
11854618	12/2022	Penney	N/A	N/A
11984148	12/2023	Penney et al.	N/A	N/A
12165687	12/2023	He et al.	N/A	N/A
12217813	12/2024	Pan	N/A	N/A
2001/0008498	12/2000	Ooishi	N/A	N/A
2002/0007476	12/2001	Kishino	N/A	N/A
2002/0078311	12/2001	Matsuzaki et al.	N/A	N/A
2002/0080677	12/2001	Watanabe et al.	N/A	N/A
2002/0181301	12/2001	Takahashi et al.	N/A	N/A
2003/0063512	12/2002	Takahashi et al.	N/A	N/A
2003/0067825	12/2002	Shimano et al.	N/A	N/A
2003/0090400	12/2002	Barker	N/A	N/A
2003/0095459	12/2002	Derner et al.	N/A	N/A
2003/0123301	12/2002	Jang et al.	N/A	N/A
2003/0193829	12/2002	Morgan et al.	N/A	N/A
2003/0231540	12/2002	Lazar et al.	N/A	N/A
2004/0004856	12/2003	Sakimura et al.	N/A	N/A
2004/0008544	12/2003	Shinozaki et al.	N/A	N/A
2004/0022093	12/2003	Lee	N/A	N/A
2004/0052142	12/2003	Ikehashi et al.	N/A	N/A
2004/0114446	12/2003	Takahashi et al.	N/A	N/A
2004/0130959	12/2003	Kawaguchi	N/A	N/A
2004/0174757	12/2003	Garverick et al.	N/A	N/A
2004/0184323	12/2003	Mori et al.	N/A	N/A
2004/0213035	12/2003	Cavaleri et al.	N/A	N/A

2004/0218431	12/2003	Chung et al.	N/A	N/A
2005/0041502	12/2004	Perner	N/A	N/A
2005/0105315	12/2004	Shin et al.	N/A	N/A
2005/0243629	12/2004	Lee	N/A	N/A
2005/0265104	12/2004	Remaklus et al.	N/A	N/A
2006/0039229	12/2005	Nakano	N/A	N/A
2006/0059196	12/2005	Sato et al.	N/A	N/A
2006/0083099	12/2005	Bae et al.	N/A	N/A
2006/0087903	12/2005	Riho et al.	N/A	N/A
2006/0119406	12/2005	Henzler et al.	N/A	N/A
2006/0176744	12/2005	Stave	N/A	N/A
2006/0262616	12/2005	Chen	N/A	N/A
2007/0008799	12/2006	Dono et al.	N/A	N/A
2007/0014174	12/2006	Ohsawa	N/A	N/A
2007/0028068	12/2006	Golding et al.	N/A	N/A
2007/0030746	12/2006	Best et al.	N/A	N/A
2007/0033339	12/2006	Best et al.	N/A	N/A
2007/0133330	12/2006	Ohsawa	N/A	N/A
2007/0153611	12/2006	Lee	N/A	N/A
2007/0230264	12/2006	Eto	N/A	N/A
2007/0237016	12/2006	Miyamoto et al.	N/A	N/A
2007/0297252	12/2006	Singh	N/A	N/A
2008/0028260	12/2007	Oyagi et al.	N/A	N/A
2008/0031068	12/2007	Yoo et al.	N/A	N/A
2008/0062742	12/2007	Wang	N/A	N/A
2008/0126893	12/2007	Harrand et al.	N/A	N/A
2008/0130394	12/2007	Dono et al.	N/A	N/A
2008/0181048	12/2007	Han	N/A	N/A
2008/0224742	12/2007	Pomichter	N/A	N/A
2008/0253212	12/2007	Iida et al.	N/A	N/A
2008/0266990	12/2007	Loeffler	N/A	N/A
2008/0288720	12/2007	Atwal et al.	N/A	N/A
2008/0301362	12/2007	Cavanna et al.	N/A	N/A
2008/0313494	12/2007	Hummler et al.	N/A	N/A
2008/0316845	12/2007	Wang et al.	N/A	N/A
2009/0021999	12/2008	Tanimura et al.	N/A	N/A
2009/0059641	12/2008	Jecideloh	N/A	N/A
2009/0077571	12/2008	Gara et al.	N/A	N/A
2009/0161457	12/2008	Wakimoto	N/A	N/A
2009/0168571	12/2008	Pyo et al.	N/A	N/A
2009/0185440	12/2008	Lee	N/A	N/A
2009/0201752	12/2008	Riho et al.	N/A	N/A
2009/0213675	12/2008	Shino	N/A	N/A
2009/0251971	12/2008	Futatsuyama	N/A	N/A
2009/0296510	12/2008	Lee et al.	N/A	N/A
2010/0005217	12/2009	Jeddeloh	N/A	N/A
2010/0005376	12/2009	Laberge et al.	N/A	N/A
2010/0054011	12/2009	Kim	N/A	N/A
2010/0074042	12/2009	Fukuda et al.	N/A	N/A
2010/0080074	12/2009	Ohmaru et al.	N/A	N/A

2010/0110809	12/2009	Kobayashi et al.	N/A	N/A
2010/0110810	12/2009	Kobayashi	N/A	N/A
2010/0131812	12/2009	Mohammad	N/A	N/A
2010/0157693	12/2009	Iwai et al.	N/A	N/A
2010/0182863	12/2009	Fukiage	N/A	N/A
2010/0260003	12/2009	Oh	N/A	N/A
2010/0329069	12/2009	Ito et al.	N/A	N/A
2011/0026290	12/2010	Noda et al.	N/A	N/A
2011/0051530	12/2010	Kushida	N/A	N/A
2011/0055495	12/2010	Remaklus, Jr. et al.	N/A	N/A
2011/0069572	12/2010	Lee et al.	N/A	N/A
2011/0122987	12/2010	Neyer	N/A	N/A
2011/0216614	12/2010	Hosoe	N/A	N/A
2011/0225355	12/2010	Kajigaya	N/A	N/A
2011/0286271	12/2010	Chen	N/A	N/A
2011/0310648	12/2010	Iwamoto et al.	N/A	N/A
2011/0317462	12/2010	Gyllenhammer et al.	N/A	N/A
2012/0014199	12/2011	Narui	N/A	N/A
2012/0059984	12/2011	Kang et al.	N/A	N/A
2012/0151131	12/2011	Kilmer et al.	N/A	N/A
2012/0213021	12/2011	Riho et al.	N/A	N/A
2012/0254472	12/2011	Ware et al.	N/A	N/A
2013/0003467	12/2012	Klein	N/A	N/A
2013/0003477	12/2012	Park et al.	N/A	N/A
2013/0057173	12/2012	Yao et al.	N/A	N/A
2013/0107623	12/2012	Kavalipurapu et al.	N/A	N/A
2013/0173971	12/2012	Zimmerman	N/A	N/A
2013/0254475	12/2012	Perego et al.	N/A	N/A
2013/0279284	12/2012	Jeong	N/A	N/A
2013/0304982	12/2012	Jung et al.	N/A	N/A
2014/0006703	12/2013	Bains et al.	N/A	N/A
2014/0006704	12/2013	Greenfield et al.	N/A	N/A
2014/0013169	12/2013	Kobla et al.	N/A	N/A
2014/0013185	12/2013	Kobla et al.	N/A	N/A
2014/0050004	12/2013	Mochida	N/A	N/A
2014/0078841	12/2013	Chopra	N/A	N/A
2014/0078842	12/2013	Oh et al.	N/A	N/A
2014/0078845	12/2013	Song	N/A	N/A
2014/0085995	12/2013	Greenfield et al.	N/A	N/A
2014/0089576	12/2013	Bains et al.	N/A	N/A
2014/0095780	12/2013	Bains et al.	N/A	N/A
2014/0095786	12/2013	Moon et al.	N/A	N/A
2014/0119091	12/2013	You et al.	N/A	N/A
2014/0136763	12/2013	Li et al.	N/A	N/A
2014/0143473	12/2013	Kim et al.	N/A	N/A
2014/0177370	12/2013	Halbert et al.	N/A	N/A
2014/0177376	12/2013	Song	N/A	N/A
2014/0189215	12/2013	Kang et al.	N/A	N/A
2014/0189228	12/2013	Greenfield et al.	N/A	N/A
2014/0219043	12/2013	Jones et al.	N/A	N/A

2014/0237307	12/2013	Kobla et al.	N/A	N/A
2014/0241099	12/2013	Seo et al.	N/A	N/A
2014/0254298	12/2013	Dally	N/A	N/A
2014/0269009	12/2013	Ghosh et al.	N/A	N/A
2014/0269021	12/2013	Yang et al.	N/A	N/A
2014/0281206	12/2013	Crawford et al.	N/A	N/A
2014/0281207	12/2013	Mandava et al.	N/A	N/A
2014/0292375	12/2013	Angelini et al.	N/A	N/A
2014/0293725	12/2013	Best et al.	N/A	N/A
2014/0317344	12/2013	Kim	N/A	N/A
2014/0355332	12/2013	Youn et al.	N/A	N/A
2014/0369109	12/2013	Lee et al.	N/A	N/A
2014/0379978	12/2013	Kim et al.	N/A	N/A
2015/0049567	12/2014	Chi	N/A	N/A
2015/0055420	12/2014	Bell et al.	N/A	N/A
2015/0078112	12/2014	Huang	N/A	N/A
2015/0089326	12/2014	Joo et al.	N/A	N/A
2015/0155027	12/2014	Abe et al.	N/A	N/A
2015/0162067	12/2014	Kim et al.	N/A	N/A
2015/0170728	12/2014	Jung et al.	N/A	N/A
2015/0199126	12/2014	Jayasena et al.	N/A	N/A
2015/0206572	12/2014	Lim et al.	N/A	N/A
2015/0213872	12/2014	Mazumder et al.	N/A	N/A
2015/0213877	12/2014	Darel	N/A	N/A
2015/0228341	12/2014	Watanabe et al.	N/A	N/A
2015/0243339	12/2014	Bell et al.	N/A	N/A
2015/0255140	12/2014	Song	N/A	N/A
2015/0262652	12/2014	Igarashi	N/A	N/A
2015/0279441	12/2014	Greenberg et al.	N/A	N/A
2015/0279442	12/2014	Hwang	N/A	N/A
2015/0294711	12/2014	Galther et al.	N/A	N/A
2015/0340077	12/2014	Akamatsu	N/A	N/A
2015/0356048	12/2014	King	N/A	N/A
2016/0019940	12/2015	Jang et al.	N/A	N/A
2016/0027498	12/2015	Ware et al.	N/A	N/A
2016/0027531	12/2015	Jones et al.	N/A	N/A
2016/0027532	12/2015	Kim	N/A	N/A
2016/0042782	12/2015	Narui et al.	N/A	N/A
2016/0078845	12/2015	Lin et al.	N/A	N/A
2016/0078911	12/2015	Fujiwara et al.	N/A	N/A
2016/0078918	12/2015	Hyun et al.	N/A	N/A
2016/0086649	12/2015	Hong et al.	N/A	N/A
2016/0086651	12/2015	Kim	N/A	N/A
2016/0093402	12/2015	Kitagawa et al.	N/A	N/A
2016/0099043	12/2015	Tu	N/A	N/A
2016/0111140	12/2015	Joo et al.	N/A	N/A
2016/0125931	12/2015	Doo et al.	N/A	N/A
2016/0133314	12/2015	Hwang et al.	N/A	N/A
2016/0140243	12/2015	Adams et al.	N/A	N/A
2016/0163372	12/2015	Lee et al.	N/A	N/A

2016/0172056	12/2015	Huh	N/A	N/A
2016/0180917	12/2015	Chishti et al.	N/A	N/A
2016/0180921	12/2015	Jeong	N/A	N/A
2016/0196863	12/2015	Shin et al.	N/A	N/A
2016/0202926	12/2015	Benedict	N/A	N/A
2016/0211008	12/2015	Benedict et al.	N/A	N/A
2016/0224262	12/2015	Mandava et al.	N/A	N/A
2016/0225433	12/2015	Bains et al.	N/A	N/A
2016/0225461	12/2015	Tuers et al.	N/A	N/A
2016/0246525	12/2015	Jeffrey et al.	N/A	N/A
2016/0336060	12/2015	Shin	N/A	N/A
2016/0343423	12/2015	Shido	N/A	N/A
2017/0011792	12/2016	Oh et al.	N/A	N/A
2017/0076779	12/2016	Bains et al.	N/A	N/A
2017/0092350	12/2016	Halbert et al.	N/A	N/A
2017/0117030	12/2016	Fisch et al.	N/A	N/A
2017/0133085	12/2016	Kim et al.	N/A	N/A
2017/0139641	12/2016	Cha et al.	N/A	N/A
2017/0140807	12/2016	Sun et al.	N/A	N/A
2017/0140811	12/2016	Joo	N/A	N/A
2017/0148504	12/2016	Saifuddin et al.	N/A	N/A
2017/0177246	12/2016	Miller et al.	N/A	N/A
2017/0186481	12/2016	Oh et al.	N/A	N/A
2017/0213586	12/2016	Kang et al.	N/A	N/A
2017/0221546	12/2016	Loh et al.	N/A	N/A
2017/0263305	12/2016	Cho	N/A	N/A
2017/0287547	12/2016	Ito et al.	N/A	N/A
2017/0323675	12/2016	Jones et al.	N/A	N/A
2017/0352399	12/2016	Yokoyama et al.	N/A	N/A
2017/0371742	12/2016	Shim et al.	N/A	N/A
2017/0372767	12/2016	Kang et al.	N/A	N/A
2018/0004592	12/2017	Bains et al.	N/A	N/A
2018/0005690	12/2017	Morgan et al.	N/A	N/A
2018/0025770	12/2017	Ito et al.	N/A	N/A
2018/0025772	12/2017	Lee et al.	N/A	N/A
2018/0060194	12/2017	Ryu et al.	N/A	N/A
2018/0061483	12/2017	Morgan	N/A	N/A
2018/0082737	12/2017	Lee	N/A	N/A
2018/0084314	12/2017	Koyama	N/A	N/A
2018/0090199	12/2017	Kim et al.	N/A	N/A
2018/0096719	12/2017	Tomishima et al.	N/A	N/A
2018/0102776	12/2017	Chandrasekar et al.	N/A	N/A
2018/0107417	12/2017	Shechter et al.	N/A	N/A
2018/0114561	12/2017	Fisch et al.	N/A	N/A
2018/0114565	12/2017	Lee	N/A	N/A
2018/0158504	12/2017	Akamatsu	N/A	N/A
2018/0158507	12/2017	Bang	N/A	N/A
2018/0182445	12/2017	Lee et al.	N/A	N/A
2018/0203621	12/2017	Ahn et al.	N/A	N/A
2018/0218767	12/2017	Wolff	N/A	N/A

2018/0261268	12/2017	Hyun et al.	N/A	N/A
2018/0294028	12/2017	Lee et al.	N/A	N/A
2018/0308539	12/2017	Ito et al.	N/A	N/A
2018/0341553	12/2017	Koudele et al.	N/A	N/A
2018/0342282	12/2017	Morgan	N/A	N/A
2018/0366182	12/2017	Hyun et al.	N/A	N/A
2019/0013059	12/2018	Akamatsu	N/A	N/A
2019/0043558	12/2018	Suh et al.	N/A	N/A
2019/0051344	12/2018	Bell et al.	N/A	N/A
2019/0066759	12/2018	Nale	N/A	N/A
2019/0066762	12/2018	Koya	N/A	N/A
2019/0088315	12/2018	Saenz et al.	N/A	N/A
2019/0088316	12/2018	Inuzuka et al.	N/A	N/A
2019/0096492	12/2018	Cai et al.	N/A	N/A
2019/0103147	12/2018	Jones et al.	N/A	N/A
2019/0130961	12/2018	Bell et al.	N/A	N/A
2019/0139599	12/2018	Ito et al.	N/A	N/A
2019/0147941	12/2018	Qin et al.	N/A	N/A
2019/0147964	12/2018	Yun et al.	N/A	N/A
2019/0161341	12/2018	Howe	N/A	N/A
2019/0172518	12/2018	Chen et al.	N/A	N/A
2019/0172521	12/2018	Morgan	N/A	N/A
2019/0187745	12/2018	Murali et al.	N/A	N/A
2019/0196730	12/2018	Imran	N/A	N/A
2019/0198078	12/2018	Hoang et al.	N/A	N/A
2019/0198090	12/2018	Lee	N/A	N/A
2019/0198099	12/2018	Mirichigni et al.	N/A	N/A
2019/0205253	12/2018	Roberts	N/A	N/A
2019/0207736	12/2018	Ben-tovim et al.	N/A	N/A
2019/0228810	12/2018	Jones et al.	N/A	N/A
2019/0228813	12/2018	Nale et al.	N/A	N/A
2019/0228815	12/2018	Morohashi et al.	N/A	N/A
2019/0237131	12/2018	Ito	N/A	N/A
2019/0237132	12/2018	Morohashi	N/A	N/A
2019/0243708	12/2018	Cha et al.	N/A	N/A
2019/0252020	12/2018	Rios et al.	N/A	N/A
2019/0267077	12/2018	Ito et al.	N/A	N/A
2019/0279706	12/2018	Kim	N/A	N/A
2019/0333573	12/2018	Shin et al.	N/A	N/A
2019/0347019	12/2018	Shin et al.	N/A	N/A
2019/0348100	12/2018	Smith et al.	N/A	N/A
2019/0348102	12/2018	Smith et al.	N/A	N/A
2019/0348103	12/2018	Jeong et al.	N/A	N/A
2019/0348107	12/2018	Shin et al.	N/A	N/A
2019/0349545	12/2018	Koh et al.	N/A	N/A
2019/0362774	12/2018	Kuramori et al.	N/A	N/A
2019/0371391	12/2018	Cha et al.	N/A	N/A
2019/0385661	12/2018	Koo et al.	N/A	N/A
2019/0385667	12/2018	Morohashi et al.	N/A	N/A
2019/0386557	12/2018	Wang et al.	N/A	N/A

2020/0005857	12/2019	Ito et al.	N/A	N/A
2020/0075106	12/2019	Tokutomi et al.	N/A	N/A
2020/0082873	12/2019	Wolff	N/A	N/A
2020/0090760	12/2019	Purahmad et al.	N/A	N/A
2020/0135263	12/2019	Brown et al.	N/A	N/A
2020/0194050	12/2019	Akamatsu	N/A	N/A
2020/0194056	12/2019	Sakurai et al.	N/A	N/A
2020/0201380	12/2019	Murali et al.	N/A	N/A
2020/0202921	12/2019	Morohashi et al.	N/A	N/A
2020/0211626	12/2019	Hiscock et al.	N/A	N/A
2020/0211633	12/2019	Okuma	N/A	N/A
2020/0211636	12/2019	Hiscock et al.	N/A	N/A
2020/0251158	12/2019	Shore et al.	N/A	N/A
2020/0294576	12/2019	Brown et al.	N/A	N/A
2020/0302994	12/2019	Enomoto et al.	N/A	N/A
2020/0321049	12/2019	Meier et al.	N/A	N/A
2020/0349995	12/2019	Shore et al.	N/A	N/A
2020/0365208	12/2019	Schreck et al.	N/A	N/A
2020/0381040	12/2019	Penney et al.	N/A	N/A
2020/0395072	12/2019	Penney et al.	N/A	N/A
2020/0411095	12/2019	Kim	N/A	N/A
2021/0005229	12/2020	Hiscock et al.	N/A	N/A
2021/0005240	12/2020	Brown	N/A	G11C 11/408
2021/0020223	12/2020	Ayyapureddi et al.	N/A	N/A
2021/0020262	12/2020	Penney et al.	N/A	N/A
2021/0026732	12/2020	Park et al.	N/A	N/A
2021/0057012	12/2020	Ayyapureddi et al.	N/A	N/A
2021/0057013	12/2020	Jenkinson et al.	N/A	N/A
2021/0057021	12/2020	Wu et al.	N/A	N/A
2021/0065755	12/2020	Kim et al.	N/A	N/A
2021/0065764	12/2020	Cheng et al.	N/A	N/A
2021/0142852	12/2020	Schreck et al.	N/A	N/A
2021/0158851	12/2020	Ayyapureddi et al.	N/A	N/A
2021/0158860	12/2020	Wu et al.	N/A	N/A
2021/0158861	12/2020	Jeong et al.	N/A	N/A
2021/0201984	12/2020	Khasawneh et al.	N/A	N/A
2021/0225432	12/2020	Enomoto et al.	N/A	N/A
2021/0241810	12/2020	Hollis et al.	N/A	N/A
2021/0265504	12/2020	Ishizu et al.	N/A	N/A
2021/0343324	12/2020	Brown et al.	N/A	N/A
2021/0350844	12/2020	Morohashi et al.	N/A	N/A
2021/0398592	12/2020	Penney et al.	N/A	N/A
2021/0407583	12/2020	Penney et al.	N/A	N/A
2022/0068348	12/2021	Bennett et al.	N/A	N/A
2022/0069992	12/2021	Ayyapureddi	N/A	N/A
2022/0165347	12/2021	Pan	N/A	N/A
2022/0230672	12/2021	Ayyapureddi et al.	N/A	N/A
2022/0293166	12/2021	Ayyapureddi et al.	N/A	N/A
2022/0415427	12/2021	Pan	N/A	N/A

2023/0047007	12/2022	Lovett	N/A	N/A
2023/0170008	12/2022	Zhang et al.	N/A	N/A
2023/0205428	12/2022	Kim et al.	N/A	N/A
2023/0206980	12/2022	He et al.	N/A	N/A
2023/0206989	12/2022	He et al.	N/A	N/A
2024/0038292	12/2023	Kim et al.	N/A	N/A
2024/0069757	12/2023	Ko et al.	N/A	N/A
2024/0135980	12/2023	Lee et al.	N/A	N/A
2024/0144988	12/2023	Seo et al.	N/A	N/A
2024/0212738	12/2023	Akamatsu et al.	N/A	N/A
2024/0427497	12/2023	Lu	N/A	N/A
2025/0068345	12/2024	Lu et al.	N/A	N/A
2025/0124963	12/2024	Lu	N/A	N/A
2025/0124964	12/2024	Lu et al.	N/A	N/A
2025/0131972	12/2024	Pan	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
1144434	12/1996	CN	N/A
1195173	12/1997	CN	N/A
1371100	12/2001	CN	N/A
101038785	12/2006	CN	N/A
101067972	12/2006	CN	N/A
101331554	12/2007	CN	N/A
101346775	12/2008	CN	N/A
101458658	12/2008	CN	N/A
101595528	12/2008	CN	N/A
101622607	12/2009	CN	N/A
102074268	12/2010	CN	N/A
102113058	12/2010	CN	N/A
102364991	12/2011	CN	N/A
102483952	12/2011	CN	N/A
103366808	12/2012	CN	N/A
103544987	12/2013	CN	N/A
103928048	12/2013	CN	N/A
104350546	12/2014	CN	N/A
105493188	12/2015	CN	N/A
105814636	12/2015	CN	N/A
106710621	12/2016	CN	N/A
107256717	12/2016	CN	N/A
107871516	12/2017	CN	N/A
108122581	12/2017	CN	N/A
108154895	12/2017	CN	N/A
109074305	12/2017	CN	N/A
109285576	12/2018	CN	N/A
109408898	12/2018	CN	N/A
109074305	12/2022	CN	N/A
H0773682	12/1994	JP	N/A
H10241385	12/1997	JP	N/A

2005-216429	12/2004	JP	N/A
2011-258259	12/2010	JP	N/A
4911510	12/2011	JP	N/A
2013-004158	12/2012	JP	N/A
20150002112	12/2014	KR	N/A
20150002783	12/2014	KR	N/A
20170058022	12/2016	KR	N/A
20180064940	12/2017	KR	N/A
1020180064940	12/2017	KR	N/A
1020180085184	12/2017	KR	N/A
20190048049	12/2018	KR	N/A
2014120477	12/2013	WO	N/A
2015030991	12/2014	WO	N/A
2017171927	12/2016	WO	N/A
2018217582	12/2017	WO	N/A
2019222960	12/2018	WO	N/A
2020010010	12/2019	WO	N/A
2020191222	12/2019	WO	N/A
2021003085	12/2020	WO	N/A
2022108808	12/2021	WO	N/A

OTHER PUBLICATIONS

US 11,264,075 B2, 03/2022, Bell et al. (withdrawn) cited by applicant
[English Abstract] Zheng, Bin , et al., “Design of Built-in DRAM for TFT-LCD Driver Chip”LCD and display, Issue 4, Aug. 15, 2009; pp. all. cited by applicant
International Application No. PCT/US20/23689, titled “Semiconductor Device Having Cam That Stores Address Signals”, dated Mar. 19, 2020, pp. all pages of application as filed. cited by applicant
U.S. Appl. No. 15/884,192 entitled ‘Semiconductor Device Performing Row Hammer Refresh Operation’ filed on Jan. 30, 2018, pp. all pages of application as filed. cited by applicant
U.S. Appl. No. 16/797,658, titles “Apparatuses and Methods for Controlling Refresh Operations”, filed Feb. 21, 2020, pp. all pages of application as filed. cited by applicant
U.S. Appl. No. 16/818,981 titled “Apparatuses and Methods for Staggered Timing of Targeted Refresh Operations” filed Mar. 13, 2020, pp. all pages of application as filed. cited by applicant
U.S. Appl. No. 16/824,460, titled “Semiconductor Device Performing Row Hammer Refresh Operation”, dated Mar. 19, 2020, pp. all pages of application as filed. cited by applicant
U.S. Appl. No. 16/025,844, titled “Apparatus and Methods for Triggering Row Hammer Address Sampling”, filed Jul. 2, 2018, pp. all pages of application as filed. cited by applicant
U.S. Appl. No. 16/783,063, titled “Apparatus and Methods for Triggering Row Hammer Address Sampling”, dated Feb. 5, 2020, pp. all pages of application as filed. cited by applicant
U.S. Appl. No. 16/805,197, titled “Apparatuses and Methods for Calculating Row Hammer Refresh Addresses in a Semiconductor Device”, dated Feb. 28, 2020, pp. all pages of application as filed. cited by applicant
U.S. Appl. No. 16/232,837, titled “Apparatuses and Methods for Distributed Targeted Refresh Operations”, filed Dec. 26, 2018, pp. all pages of application as filed. cited by applicant
U.S. Appl. No. 16/818,989, titled “Semiconductor Device Performing Row Hammer Refresh Operation”, dated Mar. 13, 2020, pp. all pages of application as filed. cited by applicant
U.S. Appl. No. 16/268,818, titled “Apparatuses and Methods for Managing Row Access Counts”, filed Feb. 6, 2019, pp. all pages of application as filed. cited by applicant
U.S. Appl. No. 16/286,187 titled “Apparatuses and Methods for Memory Mat Refresh Sequencing”

filed Feb. 26, 2019, pp. all pages of application as filed. cited by applicant

U.S. Appl. No. 16/084,119, titled “Apparatuses and Methods for Pure-Time, Self Adopt Sampling for Row Hammer Refresh Sampling”, filed Sep. 11, 2018, pp. all pages of application as filed. cited by applicant

U.S. Appl. No. 16/886,284 titled “Apparatuses and Methods for Access Based Refresh Timing” filed May 28, 2020, pp. all pages of application as filed. cited by applicant

U.S. Appl. No. 16/886,284, titled “Apparatuses and Methods for Access Based Refresh Timing”, dated May 28, 2020, pp. all pages of application as filed. cited by applicant

U.S. Appl. No. 16/358,587, titled “Semiconductor Device Having Cam That Stores Address Signals”, dated Mar. 19, 2019, pp. all pages of application as filed. cited by applicant

U.S. Appl. No. 16/375,716 titled “Apparatuses and Methods for Staggered Timing of Targeted Refresh Operations” filed Apr. 4, 2019, pp. all pages of application as filed. cited by applicant

U.S. Appl. No. 16/936,297 titled “Apparatuses and Methods for Managing Row Access Counts” filed Jul. 22, 2020, pp. all pages of application as filed. cited by applicant

U.S. Appl. No. 16/411,573 titled “Apparatuses, Systems, and Methods for a Content Addressable Memory Cell” filed May 14, 2019, pp. all pages of application as filed. cited by applicant

U.S. Appl. No. 16/428,625 titled “Apparatuses and Methods for Tracking Victim Rows” filed May 31, 2019, pp. all pages of application as filed. cited by applicant

U.S. Appl. No. 16/513,400 titled “Apparatuses and Methods for Tracking Row Accesses” filed Jul. 16, 2019, pp. all pages of application as filed. cited by applicant

U.S. Appl. No. 17/060,403 titled “Apparatuses and Methods for Adjusting Victim Data” filed Oct. 1, 2020, pp. all pages of application as filed. cited by applicant

U.S. Appl. No. 16/548,027 titled “Apparatuses, Systems, and Methods for Analog Row Access Rate Determination” filed Aug. 22, 2019, pp. all pages of application as filed. cited by applicant

U.S. Appl. No. 16/549,942 titled “Apparatuses and Methods for Lossy Row Access Counting” filed Aug. 23, 2019, pp. all pages of application as filed. cited by applicant

U.S. Appl. No. 16/546,152 titled “Apparatuses and Methods for Analog Row Access Tracking” filed Aug. 20, 2019, pp. all pages of application as filed. cited by applicant

U.S. Appl. No. 16/549,411 titled “Apparatuses and Methods for Dynamic Refresh Allocation” filed Aug. 23, 2019, pp. all pages of application as filed. cited by applicant

U.S. Appl. No. 16/655,110 titled “Apparatuses and Methods for Dynamic Targeted Refresh Steals” filed Oct. 16, 2019, pp. all pages of application as filed. cited by applicant

U.S. Appl. No. 17/154,945 titled “Apparatuses, Systems, and Methods for a Content Addressable Memory Cell” filed Jan. 21, 2021, pp. all pages of application as filed. cited by applicant

U.S. Appl. No. 17/170,616 titled “Apparatuses, Systems, and Methods for Analog Row Access Rate Determination” filed Feb. 8, 2021, pp. all pages of application as filed. cited by applicant

U.S. Appl. No. 17/168,036 titled “Apparatuses and Methods for Analog Row Access Tracking” filed Feb. 4, 2021, pp. all pages of application as filed. cited by applicant

U.S. Appl. No. 17/301,533 titled “Semiconductor Device Having Cam That Stores Address Signals” filed Apr. 6, 2021, pp. all pages of application as filed. cited by applicant

U.S. Appl. No. 17/443,056 titled “Apparatuses and Methods for Multiple Row Hammer Refresh Address Sequences” filed Jul. 20, 2021, pp. all pages of application as filed. cited by applicant

U.S. Appl. No. 17/446,710 titled “Apparatuses, Systems, and Methods for Determining Extremum Numerical values” filed Sep. 1, 2021, pp. all pages of application as filed. cited by applicant

U.S. Appl. No. 17/470,883 titled “Apparatuses and Methods for Tracking Victim Rows” filed Sep. 9, 2021, pp. all pages of application as filed. cited by applicant

U.S. Appl. No. 17/153,555 titled “Apparatuses and Methods for Dynamically Allocated Aggressor Detection” filed Jan. 20, 2021, pp. all pages of application as filed. cited by applicant

International Application No. PCT/US 19/40169 titled “Apparatus and Methods for Triggering Row Hammer Address Sampling” filed Jul. 1, 2019, pp. all pages of application as filed. cited by

applicant

International Application No. PCT/US19/64028, titled “Semiconductor Device Performing Row Hammer Refresh Operation”, dated Dec. 2, 2019, pp. all pages of application as filed. cited by applicant

International Application No. PCT/US20/26689, titled “Apparatuses and Methods for Staggered Timing of Targeted Refresh Operations”, dated Apr. 3, 2020, pp. all pages of application as filed. cited by applicant

International Application No. PCT/US20/40077, titled “Apparatuses and Methods for Monitoring Word Line Accesses”, dated Jun. 29, 2020, pp. all pages of application as filed. cited by applicant

U.S. Appl. No. 17/375,817 titled “Apparatuses and Methods for Monitoring Word Line Accesses” filed Jul. 14, 2021, pp. all pages of application as filed. cited by applicant

U.S. Appl. No. 16/788,657, titled “Semiconductor Device Performing Row Hammer Refresh Operation”, dated Feb. 12, 2020, pp. all pages of application as filed. cited by applicant

U.S. Appl. No. 15/881,256 entitled ‘Apparatuses and Methods for Detecting a Row Hammer Attack With a Bandpass Filter’ filed Jan. 26, 2018, pp. all pages of application as filed. cited by applicant

U.S. Appl. No. 17/822,033, titled, “Apparatuses and Methods for Tracking Word Line Accesses” filed Aug. 24, 2022, pp. all pages of application as filed. cited by applicant

U.S. Appl. No. 16/425,525 titled “Apparatuses And Methods For Tracking All Row Accesses” filed May 29, 2019, pp. all pages of application as filed. cited by applicant

U.S. Appl. No. 16/427,105 titled “Apparatuses And Methods For Priority Targeted Refresh Operations” filed May 30, 2019, pp. all pages of application as filed. cited by applicant

U.S. Appl. No. 16/427,140 titled “Apparatuses And Methods For Tracking Row Access Counts Between Multiple Register Stacks” filed May 30, 2019, pp. all pages of application as filed. cited by applicant

U.S. Appl. No. 16/437,811 titled “Apparatuses, Systems, And Methods For Determining Extremum Numerical Values” filed Jun. 11, 2019, pp. all pages of application as filed. cited by applicant

U.S. Appl. No. 16/537,981 titled “Apparatuses and Methods for Controlling Targeted Refresh Rates” filed Aug. 12, 2019, pp. all pages of application as filed. cited by applicant

U.S. Appl. No. 17/007,069 titled “Apparatuses and Methods for Providing Refresh Addresses” filed Aug. 31, 2020, pp. all pages of application as filed. cited by applicant

U.S. Appl. No. 17/201,941 titled “Apparatuses and Methods for Sketch Circuits for Refresh Binning” filed Mar. 15, 2021, pp. all pages of application as filed. cited by applicant

U.S. Appl. No. 17/444,925 titled “Apparatuses and Methods for Countering Memory Attacks” filed Aug. 12, 2021, pp. all pages of application as filed. cited by applicant

U.S. Appl. No. 17/456,849 titled “Apparatuses, Systems, and Methods for Main Sketch and Slim Sketch Circuitfor Row Address Tracking” filed Nov. 29, 2021, pp. all pages of application as filed. cited by applicant

U.S. Appl. No. 17/565,119 titled “Apparatuses and Methods for Row Hammer Counter Mat” filed Dec. 29, 2021, pp. all pages of application as filed. cited by applicant

U.S. Appl. No. 17/565,187 titled “Apparatuses and Methods for Row Hammer Counter Mat” filed Dec. 29, 2021, pp. all pages of application as filed. cited by applicant

U.S. Appl. No. 15/789,897, entitled “Apparatus and Methods for Refreshing Memory”, filed Oct. 20, 2017, pp. all pages of application as filed. cited by applicant

U.S. Appl. No. 15/796,340, entitled: “Apparatus and Methods for Refreshing Memory” filed Oct. 27, 2017, pp. all pages of application as filed. cited by applicant

U.S. Appl. No. 16/012,679, titled “Apparatuses and Methods for Multiple Row Hammer Refresh Address Sequences”, filed Jun. 19, 2018, pp. all pages of application as filed. cited by applicant

U.S. Appl. No. 16/020,863, titled “Semiconductor Device”, filed Jun. 27, 2018, pp. all pages of application as filed. cited by applicant

U.S. Appl. No. 16/112,471 titled "Apparatuses and Methods for Controlling Refresh Operations" filed Aug. 24, 2018, pp. all pages of application as filed. cited by applicant

U.S. Appl. No. 16/160,801, titled "Apparatuses and Methods for Selective Row Refreshes" filed Oct. 15, 2018, pp. all pages of application as filed. cited by applicant

U.S. Appl. No. 16/176,932, titled "Apparatuses and Methods for Access Based Refresh Timing", filed Oct. 31, 2018, pp. all pages of application as filed. cited by applicant

U.S. Appl. No. 16/208,217, titled "Semiconductor Device Performing Row Hammer Refresh Operation", filed Dec. 3, 2018, pp. all pages of application as filed. cited by applicant

U.S. Appl. No. 16/230,300, titled "Apparatuses and Methods for Staggered Timing of Targeted Refresh Operations" filed Dec. 21, 2018, pp. all pages of application as filed. cited by applicant

U.S. Appl. No. 16/231,327 titled "Apparatuses and Methods for Selective Row Refreshes", filed Dec. 21, 2018, pp. all pages of application as filed. cited by applicant

U.S. Appl. No. 16/237,291, titled "Apparatus and Methods for Refreshing Memory", filed Dec. 31, 2018, pp. all pages of application as filed. cited by applicant

U.S. Appl. No. 16/290,730, titled "Semiconductor Device Performing Row Hammer Refresh Operation", filed Mar. 1, 2019, pp. all pages of application as filed. cited by applicant

U.S. Appl. No. 16/374,623, titled "Semiconductor Device Performing Row Hammer Refresh Operation", filed Apr. 3, 2019, pp. all pages of application as filed. cited by applicant

U.S. Appl. No. 16/411,698 title "Semiconductor Device" filed May 14, 2019, pp. all pages of application as filed. cited by applicant

U.S. Appl. No. 16/427,330 titled "Apparatuses and Methods for Storing Victim Row Data" filed May 30, 2019, pp. all pages of application as filed. cited by applicant

U.S. Appl. No. 16/431,641 titled "Apparatuses and Methods for Controlling Steal Rates" filed Jun. 4, 2019, pp. all pages of application as filed. cited by applicant

U.S. Appl. No. 16/682,606, titled "Apparatuses And Methods For Distributing Row Hammer Refresh Events Across A Memory Device", filed Nov. 13, 2019, pp. all pages of application as filed. cited by applicant

U.S. Appl. No. 17/102,266, titled "Apparatuses and Methods for Tracking Word Line Accesses", dated Nov. 23, 2020, pp. all pages of application as filed. cited by applicant

U.S. Appl. No. 15/876,566 entitled 'Apparatuses and Methods for Calculating Row Hammer Refresh Addresses in a Semiconductor Device' filed Jan. 22, 2018, pp. all pages of application as filed. cited by applicant

International Application No. PCT/US20/32684, titled "Apparatuses, Systems, and Methods for a Content Addressable Memory Cell", dated May 13, 2020, pp. all pages of application as filed. cited by applicant

U.S. Appl. No. 15/656,084, titled "Apparatuses And Methods For Targeted Refreshing Of Memory", filed Jul. 21, 2017, pp. all pages of application as filed. cited by applicant

U.S. Appl. No. 16/459,520 titled "Apparatuses and Methods for Monitoring Word Line Accesses", filed Jul. 1, 2019, pp. all pages of application as filed. cited by applicant

International Application No. PCT/US18/55821 "Apparatus and Methods for Refreshing Memory" filed Oct. 15, 2018, pp. all pages of application as filed. cited by applicant

U.S. Appl. No. 15/715,846, entitled "Semiconductor Device", filed Sep. 26, 2017, pp. all pages of application as filed. cited by applicant

U.S. Appl. No. 15/888,993, entitled "Apparatuses and Methods for Controlling Refresh Operations", filed Feb. 5, 2018, pp. all pages of application as filed. cited by applicant

U.S. Appl. No. 16/190,627 titled "Apparatuses and Methods for Targeted Refreshing of Memory" filed Nov. 14, 2018, pp. all pages of application as filed. cited by applicant

U.S. Appl. No. 16/459,507 titled "Apparatuses and Methods for Adjusting Victim Data", filed Jul. 1, 2019, pp. all pages of application as filed. cited by applicant

U.S. Appl. No. 15/281,818, entitled: "Semiconductor Device" filed Sep. 30, 2016, pp. all pages of

application as filed. cited by applicant
Kim, et al., “Flipping Bits in Memory Without Accessing Them: An Experimental Study of DRAM Disturbance Errors”, IEEE, Jun. 2014, 12 pgs. cited by applicant
Stout, Thomas et al., “Voltage Source Based Voltage-to-Time Converter”, IEEE, downloaded Jul. 2020, p. All. cited by applicant
U.S. Appl. No. 18/741,485 titled “Apparatuses and Methods for Aggressor Queue Based Mitigation Threshold” filed Jun. 12, 2024, pp. all pages of the application as filed. cited by applicant
U.S. Appl. No. 18/743,783 titled “Apparatuses and Methods for Per-Row Count Based Refresh Target Identification” filed Jun. 14, 2024, pp. all pages of the application as filed. cited by applicant
U.S. Appl. No. 18/744,988 titled “Apparatuses and Methods for Per-Row Count Based Refresh Target Identification With Sorting” filed Jun. 17, 2024, pp. all pages of the application as filed. cited by applicant
U.S. Appl. No. 18/745,068 titled “Apparatuses and Methods for Per Row Activation Counter Testing” filed Jun. 17, 2024, pp. all pages of the application as filed. cited by applicant

Primary Examiner: Yoha; Connie C

Attorney, Agent or Firm: Dorsey & Whitney LLP

Background/Summary

CROSS REFERENCE TO RELATED APPLICATION(S) (1) This application is a continuation of U.S. patent application Ser. No. 17/153,555 filed Jan. 20, 2021 and issued as U.S. Pat. No. 11,482,275 on Oct. 25, 2022. The aforementioned application, and issued patent, is incorporated herein by reference, in its entirety, for any purpose.

BACKGROUND

(1) This disclosure relates generally to semiconductor devices, and more specifically to semiconductor memory devices. In particular, the disclosure relates to volatile memory, such as dynamic random access memory (DRAM). Information may be stored on individual memory cells of the memory as a physical signal (e.g., a charge on a capacitive element). The memory may be a volatile memory, and the physical signal may decay over time (which may degrade or destroy the information stored in the memory cells). It may be necessary to periodically refresh the information in the memory cells by, for example, rewriting the information to restore the physical signal to an initial value.

(2) As memory components have decreased in size, the density of memory cells has greatly increased. Various access patterns to a particular memory cell or group of memory cells (often referred to as an attack) may cause an increased rate of data degradation in nearby memory cells. Memory cells affected by the attack may be identified and refreshed as part of a targeted refresh operation. The memory may track access patterns to various memory addresses in order to determine if they involved in an attack. However, it may be extremely storage intensive to track accesses to every address.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a block diagram of a semiconductor device according an embodiment of the disclosure.

(2) FIG. 2 is a block diagram of a refresh control circuit according to an embodiment of the present disclosure.

(3) FIG. 3 is a block diagram of a portion of a memory according to some embodiments of the present disclosure.

(4) FIG. 4 is a block diagram of an aggressor detector according to some embodiments of the present disclosure.

(5) FIG. 5 is a block diagram of a method according to some embodiments of the present disclosure.

DETAILED DESCRIPTION

(6) The following description of certain embodiments is merely exemplary in nature and is in no way intended to limit the scope of the disclosure or its applications or uses. In the following detailed description of embodiments of the present systems and methods, reference is made to the accompanying drawings which form a part hereof, and which are shown by way of illustration specific embodiments in which the described systems and methods may be practiced. These embodiments are described in sufficient detail to enable those skilled in the art to practice presently disclosed systems and methods, and it is to be understood that other embodiments may be utilized and that structural and logical changes may be made without departing from the spirit and scope of the disclosure. Moreover, for the purpose of clarity, detailed descriptions of certain features will not be discussed when they would be apparent to those with skill in the art so as not to obscure the description of embodiments of the disclosure. The following detailed description is therefore not to be taken in a limiting sense, and the scope of the disclosure is defined only by the appended claims.

(7) Information in a volatile memory device may be stored in memory cells (e.g., as a charge on a capacitive element), and may decay over time. The memory cells may be organized into rows (wordlines) and columns (bit lines), in each bank of a memory array. The memory cells may be refreshed on a row-by-row basis. In order to prevent information from being lost or corrupted due to this decay, the memory may carry out a background refresh process, such as auto-refresh operations as part of a self-refresh mode. During a refresh operation, information may be rewritten to the wordline to restore its initial state. The auto-refresh operations may be performed on the wordlines of the memory in a sequence such that over time the wordlines of the memory are refreshed at a rate faster than the expected rate of data degradation.

(8) Various attack patterns, such as repeated access to a particular row of memory (e.g., an aggressor row) may cause an increased rate of decay in neighboring rows (e.g., victim rows) due, for example, to electromagnetic coupling between the rows. The pattern of repeated accesses may be referred to as a 'row hammer'. These repeated accesses may be part of a deliberate attack against the memory and/or may be due to 'natural' access patterns of the memory. The increased rate of decay in the victim rows may require that they be refreshed as part of a targeted refresh operation to prevent information from being lost.

(9) The memory may track accesses to different rows to determine if those rows are aggressors or not. An aggressor detector circuit may store row potential aggressor addresses (e.g., row addresses which were previously accessed) and may compare those stored row addresses to a currently accessed row address. The ability of the memory to accurately catch aggressor addresses may depend, in part on how the memory stores potential aggressor addresses. Some solutions may include a storage structure for storing potential aggressor addresses for each bank of the memory. However, this solution may not scale well as the number of banks increases. Further, it may be inefficient to divide the storage on a bank-by-bank basis, as it is unlikely that every bank will be attacked at the same time. There may thus be a need to increase the efficiency of aggressor address storage and tracking.

(10) The present disclosure is drawn to apparatuses, systems, and methods for dynamically allocated aggressor detection. A memory device may have an aggressor address storage structure which is shared between one or more banks of the memory. The individual storage slots of the

storage structure may be dynamically allocated between different banks (e.g., based on the accesses to those banks). For example, the storage structure may store row addresses along with their associated bank addresses. The aggressors may be determined based on both the stored row and bank addresses. Once an aggressor is detected, its victims may be located and refreshed based on both the row and bank address. Since the slots in the storage structure are not permanently assigned to a given bank, the space may be dynamically allocated to different banks based on the access patterns to those banks. In this manner, if a single bank is attacked, more storage may be available for tracking that attack, even though the aggressor address storage may include fewer total storage spaces than may be used in bank specific solutions. In some embodiments, the shared aggressor address storage may also move to an area of the memory die which is more distant from the banks (e.g., not in the bank logic region) which may help reduce free up space closer to the banks. In some embodiments, the sharing of aggressor storage may also allow for a reduction in the total amount of aggressor storage on the memory device compared to memory devices which have a separate aggressor storage for each bank.

(11) FIG. 1 is a block diagram of a semiconductor device according an embodiment of the disclosure. The semiconductor device **100** may be a semiconductor memory device, such as a DRAM device integrated on a single semiconductor chip.

(12) The semiconductor device **100** includes a memory array **118**. The memory array **118** is shown as including a plurality of memory banks. In the embodiment of FIG. 1, the memory array **118** is shown as including eight memory banks BANK0-BANK7. More or fewer banks may be included in the memory array **118** of other embodiments. Each memory bank includes a plurality of word lines WL, a plurality of bit lines BL and/BL, and a plurality of memory cells MC arranged at intersections of the plurality of word lines WL and the plurality of bit lines BL and/BL. The selection of the word line WL is performed by a row decoder **108** and the selection of the bit lines BL and/BL is performed by a column decoder **110**. In the embodiment of FIG. 1, the row decoder **108** includes a respective row decoder for each memory bank and the column decoder **110** includes a respective column decoder for each memory bank. The bit lines BL and/BL are coupled to a respective sense amplifier (SAMP). Read data from the bit line BL or/BL is amplified by the sense amplifier SAMP, and transferred to read/write amplifiers **120** over complementary local data lines (LIOT/B), transfer gate (TG), and complementary main data lines (MIOT/B). Conversely, write data outputted from the read/write amplifiers **120** is transferred to the sense amplifier SAMP over the complementary main data lines MIOT/B, the transfer gate TG, and the complementary local data lines LIOT/B, and written in the memory cell MC coupled to the bit line BL or/BL.

(13) The semiconductor device **100** may employ a plurality of external terminals that include command and address (C/A) terminals coupled to a command and address bus to receive commands and addresses, and a CS signal, clock terminals to receive clocks CK and/CK, data terminals DQ to provide data, and power supply terminals to receive power supply potentials VDD, VSS, VDDQ, and VSSQ.

(14) The clock terminals are supplied with external clocks CK and/CK that are provided to an input circuit **112**. The external clocks may be complementary. The input circuit **112** generates an internal clock ICLK based on the CK and/CK clocks. The ICLK clock is provided to the command decoder **110** and to an internal clock generator **114**. The internal clock generator **114** provides various internal clocks LCLK based on the ICLK clock. The LCLK clocks may be used for timing operation of various internal circuits. The internal data clocks LCLK are provided to the input/output circuit **122** to time operation of circuits included in the input/output circuit **122**, for example, to data receivers to time the receipt of write data.

(15) The C/A terminals may be supplied with memory addresses. The memory addresses supplied to the C/A terminals are transferred, via a command/address input circuit **102**, to an address decoder **104**. The address decoder **104** receives the address and supplies a decoded row address XADD to the row decoder **108** and supplies a decoded column address YADD to the column

decoder **110**. The address decoder **104** may also supply a decoded bank address BADD, which may indicate the bank of the memory array **118** containing the decoded row address XADD and column address YADD. The C/A terminals may be supplied with commands. Examples of commands include timing commands for controlling the timing of various operations, access commands for accessing the memory, such as read commands for performing read operations and write commands for performing write operations, as well as other commands and operations. The access commands may be associated with one or more row address XADD, column address YADD, and bank address BADD to indicate the memory cell(s) to be accessed.

(16) The commands may be provided as internal command signals to a command decoder **106** via the command/address input circuit **102**. The command decoder **106** includes circuits to decode the internal command signals to generate various internal signals and commands for performing operations. For example, the command decoder **106** may provide a row command signal to select a word line and a column command signal to select a bit line.

(17) The device **100** may receive an access command which is a read command. When a read command is received, and a bank address, a row address and a column address are timely supplied with the read command, read data is read from memory cells in the memory array **118** corresponding to the row address and column address. The read command is received by the command decoder **106**, which provides internal commands so that read data from the memory array **118** is provided to the read/write amplifiers **120**. The read data is output to outside from the data terminals DQ via the input/output circuit **122**.

(18) The device **100** may receive an access command which is a write command. When the write command is received, and a bank address, a row address and a column address are timely supplied with the write command, write data supplied to the data terminals DQ is written to a memory cells in the memory array **118** corresponding to the row address and column address. The write command is received by the command decoder **106**, which provides internal commands so that the write data is received by data receivers in the input/output circuit **122**. Write clocks may also be provided to the external clock terminals for timing the receipt of the write data by the data receivers of the input/output circuit **122**. The write data is supplied via the input/output circuit **122** to the read/write amplifiers **120**, and by the read/write amplifiers **120** to the memory array **118** to be written into the memory cell MC.

(19) The device **100** may also receive commands causing it to carry out one or more refresh operations as part of a self-refresh mode. In some embodiments, the self-refresh mode command may be externally issued to the memory device **100**. In some embodiments, the self-refresh mode command may be periodically generated by a component of the device. In some embodiments, when an external signal indicates a self-refresh entry command, the refresh signal AREF may also be activated. The refresh signal AREF may be a pulse signal which is activated when the command decoder **106** receives a signal which indicates entry to the self-refresh mode. The refresh signal AREF may be activated once immediately after command input, and thereafter may be cyclically activated at desired internal timing. The refresh signal AREF may be used to control the timing of refresh operations during the self-refresh mode. Thus, refresh operations may continue automatically. A self-refresh exit command may cause the automatic activation of the refresh signal AREF to stop and may cause the device **100** to return to an idle state and/or resume other operations.

(20) The refresh signal AREF is supplied to the refresh control circuit **116**. The refresh control circuit **116** supplies a refresh row address RXADD to the row decoder **108**, which may refresh one or more wordlines WL indicated by the refresh row address RXADD. In some embodiments, the refresh address RXADD may represent a single wordline. In some embodiments, the refresh address RXADD may represent multiple wordlines, which may be refreshed sequentially or simultaneously by the row decoder **108**. In some embodiments, the number of wordlines represented by the refresh address RXADD may vary from one refresh address to another. The

refresh control circuit **116** may control a timing of the refresh operation, and may generate and provide the refresh address RXADD. The refresh control circuit **116** may be controlled to change details of the refreshing address RXADD (e.g., how the refresh address is calculated, the timing of the refresh addresses, the number of wordlines represented by the address), or may operate based on internal logic.

(21) The refresh control circuit **116** may selectively output a targeted refresh address (e.g., which specifies one or more victim address based on an aggressor) or an automatic refresh address (e.g., from a sequence of auto-refresh addresses) as the refresh address RXADD. Based on the type of refresh address RXADD (and in some embodiments, one more additional signals indicating the type of operation), the row decoder **108** may perform a targeted refresh or auto-refresh operation. The automatic refresh addresses may be from a sequence of addresses which are provided based on activations of the refresh signal AREF. The refresh control circuit **116** may cycle through the sequence of auto-refresh addresses at a rate determined by AREF. In some embodiments, the auto-refresh operations may generally occur with a timing such that the sequence of auto-refresh addresses is cycled such that no information is expected to degrade in the time between auto-refresh operations for a given wordline. In other words, auto-refresh operations may be performed such that each wordline is refreshed at a rate faster than the expected rate of information decay.

(22) The refresh control circuit **116** may also determine targeted refresh addresses which are addresses that require refreshing (e.g., victim addresses corresponding to victim rows) based on the access pattern of nearby addresses (e.g., aggressor addresses corresponding to aggressor rows) in the memory array **118**. The refresh control circuit **116** may use one or more signals of the device **100** to calculate the targeted refresh address RXADD. For example, the refresh address RXADD may be calculated based on the row addresses XADD provided by the address decoder.

(23) In some embodiments, the refresh control circuit **116** may sample the current value of the row address XADD provided by the address decoder **104** along a row address bus, and determine a targeted refresh address based on one or more of the sampled addresses. The sampled addresses may be stored in a data storage unit of the refresh control circuit. When a row address XADD is sampled, it may be compared to the stored addresses in the data storage unit. In some embodiments, the aggressor address may be determined based on the sampled and/or stored addresses. For example, the comparison between the sampled address and the stored addresses may be used to update a count value (e.g., an access count) associated with the stored addresses and the aggressor address may be calculated based on the count values. The refresh addresses RXADD may then be used based on the aggressor addresses.

(24) While in general the present disclosure refers to determining aggressor and victim wordlines and addresses, it should be understood that as used herein, an aggressor wordline does not necessarily need to cause data degradation in neighboring wordlines, and a victim wordline does not necessarily need to be subject to such degradation. The refresh control circuit **116** may use some criteria to judge whether an address is an aggressor address, which may capture potential aggressor addresses rather than definitively determining which addresses are causing data degradation in nearby victims. For example, the refresh control circuit **116** may determine potential aggressor addresses based on a pattern of accesses to the addresses and this criteria may include some addresses which are not aggressors, and miss some addresses which are. Similarly, victim addresses may be determined based on which wordlines are expected to be effected by aggressors, rather than a definitive determination of which wordlines are undergoing an increased rate of data decay.

(25) As described in more detail herein, the refresh control circuit **116** may be divided between components which are specific to a given bank and components which are shared between banks. The aggressor detector portion of the refresh control circuit **116** may be common between one or more banks, while portions of the refresh control circuit **116** which generate the refresh address RXADD may be bank specific. Accordingly, there may be a single aggressor detector portion,

while there may be multiple refresh address generator portions (e.g., one for every bank). In some embodiments, these components may be placed in different parts of the physical die which holds the memory device. An example refresh control circuit is described in more detail in FIG. 2.

(26) The power supply terminals are supplied with power supply potentials VDD and VSS. The power supply potentials VDD and VSS are supplied to an internal voltage generator circuit **124**. The internal voltage generator circuit **124** generates various internal potentials VPP, VOD, VARY, VPERI, and the like based on the power supply potentials VDD and VSS supplied to the power supply terminals. The internal potential VPP is mainly used in the row decoder **108**, the internal potentials VOD and VARY are mainly used in the sense amplifiers SAMP included in the memory array **118**, and the internal potential VPERI is used in many peripheral circuit blocks.

(27) The power supply terminals are also supplied with power supply potentials VDDQ and VSSQ. The power supply potentials VDDQ and VSSQ are supplied to the input/output circuit **122**. The power supply potentials VDDQ and VSSQ supplied to the power supply terminals may be the same potentials as the power supply potentials VDD and VSS supplied to the power supply terminals in an embodiment of the disclosure. The power supply potentials VDDQ and VSSQ supplied to the power supply terminals may be different potentials from the power supply potentials VDD and VSS supplied to the power supply terminals in another embodiment of the disclosure. The power supply potentials VDDQ and VSSQ supplied to the power supply terminals are used for the input/output circuit **122** so that power supply noise generated by the input/output circuit **122** does not propagate to the other circuit blocks.

(28) FIG. 2 is a block diagram of a refresh control circuit according to an embodiment of the present disclosure. The refresh control circuit **216** may, in some embodiments, be included in the refresh control circuit **116** of FIG. 1. Certain internal components and signals of the refresh control circuit **216** are shown to illustrate the operation of the refresh control circuit **216**. The dotted line **218** is shown to represent that in certain embodiments, each of the components (e.g., the RHR state control circuit **236**, the refresh address generator **234**, the local aggressor storage **238**, and the row decoder **208**) may correspond to a particular bank of memory, and that these components may be repeated for each of the banks of memory. Similarly, other components, such as the sample timing circuit **230** and aggressor detector **232** may be shared amongst the banks. For the sake of brevity, since the components repeated for each bank may be generally similar to each other, only the interaction of the shared components with a single set of bank-by-bank components will be described in detail.

(29) A DRAM interface **226** may provide one or more signals to an address refresh control circuit **216** and row decoder **208**. The refresh control circuit **216** may include a sample timing circuit **230**, an aggressor detector circuit **232**, a row hammer refresh (RHR) state control circuit **236** and a refresh address generator **234**. The DRAM interface **226** may provide one or more control signals, such as a refresh signal AREF, and a row address XADD. The refresh control circuit **216** provides refresh address RXADD with timing based on the refresh signal AREF, wherein some of the refresh addresses are based on the received row address XADD.

(30) In some embodiments, the aggressor detector circuit **232** may sample the current row address XADD responsive to an activation a sampling signal ArmSample. The aggressor detector circuit **232** may be coupled to the row addresses XADD and bank addresses BADD along the address bus, but may only receive (e.g., process, pay attention to) the current value of the row address XADD and bank address BADD when there is an activation of the sampling signal ArmSample. The sampled addresses may be stored in the aggressor circuit **232** and/or compared to previously stored addresses. The aggressor detector circuit **232** may provide a match address HitXADD (e.g., an identified aggressor address) based on a currently sampled row address XADD and bank address BADD and/or previously sampled row addresses and bank addresses. The aggressor address HitXADD may be a sampled/stored row address and may be directed to bank level circuits (e.g., local aggressor storage **238**) based on the bank address BADD which was sampled/stored along

with the sampled stored row address.

(31) The RHR state control circuit **236** may provide the signal RHR to indicate that a row hammer refresh (e.g., a refresh of the victim rows corresponding to an identified aggressor row) should occur. The RHR state control circuit **236** may also provide an internal refresh signal IREF, to indicate that an auto-refresh should occur. Responsive to an activation of RHR or IREF, the refresh address generator **234** may provide a refresh address RXADD, which may be an auto-refresh address or may be one or more victim addresses corresponding to victim rows of the aggressor row corresponding to the match address HitXADD stored in the local aggressor storage **238** (or directly provided by the aggressor detector circuit **232**). The RHR state control circuit **236** may provide a set of activations of RHR and IREF responsive to the refresh signal AREF, representing a number of refresh pumps to each activation of the refresh signal AREF. The row decoder **208** may perform a targeted refresh operation responsive to the refresh address RXADD and the row hammer refresh signal RHR. The row decoder **208** may perform an auto-refresh operation based on the refresh address RXADD and the internal refresh signal IREF.

(32) The DRAM interface **226** may represent one or more components which provides signals to components of the bank. In some embodiments, the DRAM interface **226** may represent a memory controller coupled to the semiconductor memory device (e.g., device **100** of FIG. 1). In some embodiments, the DRAM interface **226** may represent components such as the command address input circuit **102**, the address decoder **104**, and/or the command decoder **106** of FIG. 1. The DRAM interface **226** may provide a row address XADD, a bank address BADD the refresh signal AREF, and access signals such as an activation signal ACT and a pre-charge signal PRE. The refresh signal AREF may be a periodic signal which may indicate when an auto-refresh operation is to occur. The access signals ACT and PRE may generally be provided as part of an access operation along with a row address XADD and bank address BADD. The activation signal ACT may be provided to activate a bank and row of the memory associated with the associated bank and row address. The pre-charge signal PRE may be provided to pre-charge the bank and row of the memory specified by the bank and row address.

(33) The row address XADD may be a signal including multiple bits (which may be transmitted in series or in parallel) and may correspond to a specific row of an activated memory bank. Similarly, the bank address BADD may be a multi-bit signal which corresponds to a specific bank of the memory array. The number of bits of the row and bank address may be based on a number of banks, and a number of rows in each bank. For example, the row address may be 17 bits long, while the bank address may be 5 bits long.

(34) In the example embodiment of FIG. 2, the refresh control circuit **216** uses sampling to monitor a portion of the addresses XADD and BADD provided along the address bus. Accordingly, instead of responding to every address, the refresh control circuit **216** may sample the current value of the address XADD and BADD on the address bus, and may determine which addresses are aggressors based on the sampled addresses. The timing of sampling by the refresh control circuit **216** may be controlled by the sample timing circuit **230** which provides the sampling signal ArmSample. The sample timing circuit **230** may provide activations of the sampling signal ArmSample, and each activation of the signal ArmSample may indicate that a current value of the row address should be sampled. An activation of ArmSample may be a 'pulse', where ArmSample is raised to a high logic level and then returns to a low logic level. The activations of the signal ArmSample may be provided with periodic timing, random timing, semi-random timing, pseudo-random timing, or combinations thereof. In other embodiments, sampling may not be used, and the aggressor detector circuit **232** may receive every value of the row address XADD and bank address BADD along the row address bus.

(35) As described in more detail herein, the aggressor detector circuit **232** may determine aggressor addresses based on one or more of the sampled row and bank addresses, and then may provide the determined aggressor address as the match address HitXADD. Although based on a row and bank

address pair, in some embodiments, the match address HitXADD may represent just an identified aggressor row address, while the bank address may be used to route that match address HitXADD to the proper bank portion **218**. The aggressor detector circuit **232** may include a data storage unit (e.g., a number of registers), which may be used to store sampled row and bank addresses. When the aggressor detector circuit **232** samples a new value of the row address XADD and bank address BADD (e.g., responsive to an activation of ArmSample) it may compare the sampled row and bank address to the row/bank addresses stored in the data storage unit. In some embodiments, the match address HitXADD may be one of the addresses stored in the aggressor detector circuit **232** which has been matched by the sampled addresses the most frequently.

(36) The RHR state control circuit **236** may receive the refresh signal AREF and provide the auto-refresh signal IREF and the row hammer refresh signal RHR. The refresh signal AREF may be periodically generated and may be used to control the timing of refresh operations. The memory device may carry out a sequence of auto-refresh operations in order to periodically refresh the rows of the memory device. The RHR signal may be generated in order to indicate that the device should refresh a particular targeted row (e.g., a victim row) instead of an address from the sequence of auto-refresh addresses. The RHR state control circuit **236** may also provide an internal refresh signal IREF, which may indicate that an auto-refresh operation should take place. In some embodiments, the signals RHR and IREF may be generated such that they are not active at the same time (e.g., are not both at a high logic level at the same time). In some embodiments, IREF may be activated for every refresh operation, and an auto-refresh operation may be performed unless RHR is also active, in which case a targeted refresh operation is performed instead.

(37) In some embodiments, the refresh control circuit **216** may perform multiple refresh operations responsive to each activation of the refresh signal AREF. For example, each time the refresh signal AREF is received, the refresh control circuit **216** may perform N different refresh operations, by providing N different refresh addresses RXADD. Each refresh operation may be referred to as a 'pump'. The different pumps generated in response to the refresh signal AREF may be a mix of auto-refresh and targeted refresh operations. For example, if 4 pumps are generated, two may be used for auto-refresh operations and two may be used for targeted refresh operations. Other patterns may be used in other embodiments. In some embodiments, the pattern of targeted and auto-refresh operations may vary between different groups of pumps.

(38) The refresh address generator **234** may receive the row hammer refresh signal RHR and the match address HitXADD. The match address HitXADD may represent an aggressor row. The refresh address generator **234** may determine the locations of one or more victim rows based on the match address HitXADD and provide them as the refresh address RXADD when the signal RHR indicates a targeted refresh operation. In some embodiments, the victim rows may include rows which are physically adjacent to the aggressor row (e.g., HitXADD+1 and HitXADD-1). In some embodiments, the victim rows may also include rows which are physically adjacent to the physically adjacent rows of the aggressor row (e.g., HitXADD+2 and HitXADD-2). Other relationships between victim rows and the identified aggressor rows may be used in other examples. For example, +/-3, +/-4, and/or other rows may also be refreshed.

(39) The refresh address generator **234** may determine the value of the refresh address RXADD based on the row hammer refresh signal RHR. In some embodiments, when the signal RHR is not active, the refresh address generator **234** may provide one of a sequence of auto refresh addresses. When the signal RHR is active, the refresh address generator **234** may provide a targeted refresh address, such as a victim address, as the refresh address RXADD. In some embodiments, the refresh address generator **234** may count activations of the signal RHR, and may provide closer victim rows (e.g., HitXADD+/-1) more frequently than victim rows which are further away from the aggressor address (e.g., HitXADD+/-2).

(40) In some embodiments, the match address HitXADD may be stored in an optional local aggressor storage **238**. While the refresh address generator **234** may, in some embodiments, retrieve

the match address HitXADD directly from the aggressor detector circuit **232**. However, this may lead to timing difficulties, for example if the aggressor detector circuit **232** is located in a portion of the memory device which is not proximal to the memory bank components **218**. Accordingly, when a match address HitXADD is identified, it may be stored in a local aggressor storage **238**. The aggressor detector circuit **232** may provide the match address HitXADD to the appropriate local aggressor storage **238** based on the bank address associated with the match address (e.g., the bank address which was received/stored with the match address). In some embodiments, the match address HitXADD may be provided to the local aggressor storage **238** along the address bus (e.g., a row address bus). Various timing logic may be used to prevent conflicts with other addresses along the address bus. In some embodiments, the aggressor detector circuit **232** and local aggressor storage **238** may be coupled by a dedicated bus (e.g., a different bus from the address bus used to carry the row and bank addresses as part of an access operation). The dedicated bus may operate in a serial fashion, a parallel fashion, or combinations thereof.

(41) The row decoder **208** may perform one or more operations on the memory array (not shown) based on the received signals and addresses. For example, responsive to the activation signal ACT and the row address XADD (and IREF and RHR being at a low logic level), the row decoder **208** may direct one or more access operations (for example, a read operation) on the specified row address XADD. Responsive to the RHR signal being active, the row decoder **208** may refresh the refresh address RXADD.

(42) FIG. **3** is a block diagram of a portion of a memory according to some embodiments of the present disclosure. The memory **300** may, in some embodiments, represent a portion of the memory **102** of FIG. **1**. In particular, the memory **300** shows certain components useful for discussing the operation of an aggressor detector circuit **302**. The aggressor detector circuit **302** may, in some embodiments, be included in the aggressor detector circuit **232** of FIG. **2**.

(43) The aggressor detector circuit **302** includes an address storage structure **304** and storage logic **306** which manages the information stored in the address storage structure **304**. The address storage structure **304** has a number of individual slots (e.g., rows as illustrated in FIG. **3**), each of which stores one or more associated pieces of information. For example, in FIG. **3**, each slot of the address storage structure **304** holds a row address XADD, a bank address BADD, and a count value Count. As discussed in more detail herein, other embodiments may store different information and/or may store information in different ways.

(44) The storage logic **306** may represent one or more components which manage the contents of the address storage structure **304**. When the sampling signal ArmSample is provided by the sample timing logic **308**, the storage logic **306** may capture the next row address XADD and bank address BADD along the address bus. The storage logic **306** may compare the received row and bank address to row and bank addresses stored in the address storage structure **304**. The storage logic **306** may determine if the received row and bank address match one of the stored row and bank addresses.

(45) In some embodiments, the address storage structure **304** may include content addressable memory (CAM) cells. Each CAM cell may store an individual bit of information. For example, if the row address is i bits long and the bank address is j bits long, then each slot may include $i+j$ CAM cells. The CAM cells which make the address storage portion of a slot may act together to provide a match signal which indicates if all of the bits of received information (e.g., a sampled row/bank address) matches the bits of the stored information (e.g., a stored row/bank address). For example, each CAM cell may provide a cell match signal, and the cell match signals may be logically combined (e.g. with AND logic) to determine an overall match signal. In some embodiments, the match signal may only be provided if all of the bits of the sampled bank address match the stored bank address and all of the bits of the sampled row address associated with the sampled bank address match the stored row address associated with the stored bank address.

(46) The aggressor detector **302** may use a count based scheme to determine if a sampled row and

bank address are an aggressor address. Accordingly, if there is a match between the received (e.g., sampled) row and bank address and one of the stored row and bank addresses, then the count value associated with the stored row and bank address may be changed (e.g., incremented). The storage logic **306** may include one or more count logic circuits. Responsive to a match signal from the aggressor address storage **304**, the count value in the slot which provided the match signal may be read out, and updated (e.g., incremented).

(47) The updated count value may be compared to a threshold by a comparator circuit of the storage logic **306**. Based on that comparison (e.g., if the updated count is greater than the threshold), the storage logic may determine if the sampled bank/row address are aggressors, and may provide the sampled row address as the match address HitXADD and the sampled bank address BADD as the match address HitBADD. If the sampled address is not a match address (e.g., if the count is below the threshold) then the updated count value may be written back to the aggressor address storage **304**. If the sampled address is provided as the match address HitXADD/HitBADD, the count value may be further changed (e.g., decremented by the threshold value, reset to an initial value such as 0, etc.) before being written back to the aggressor address storage **304**. In some embodiments, the stored row and bank address may be removed from the address storage structure **304** once they are used to provide the match address HitXADD and HitBADD.

(48) If there is not a match between the received row and bank address and any of the stored row and bank addresses, the storage logic **306** may store the received row and bank address in the address storage structure **304**. The storage logic **306** may determine if there is open space (e.g., a slot which is not currently in use) in the aggressor storage structure **304**, and if so, store the received row and bank address in the open space. If there is not an open space, then the storage logic **306** may use one or more criterion to determine whether to and where to store the new row and bank addresses. For example, the stored row and bank address associated with a lowest of the count values may be replaced.

(49) In some embodiments, the storage logic **306** may use different criterion for determining which address is a match address HitXADD and HitBADD. For example, the storage logic **306** may compare the received row and bank address to the stored row and bank addresses, and provide the received row and bank address as the address HitXADD and HitBADD if there is a match. In such embodiments, the aggressor storage structure **304** may not include count values. In another example, the storage logic **306** may provide the stored row and bank address which have the highest count value as the address HitXADD and HitBADD. Other schemes for identifying the aggressor address so it can be provided as the match address HitXADD and HitBADD may be used in other example embodiments.

(50) The match row and bank address HitXADD and HitBADD respectively may be provided to the refresh circuitry specific to the bank associated with the bank match address HitBADD. The match bank address HitBADD may be used to route the match row address HitXADD to the circuitry specific to the bank associated with the bank address HitBADD.

(51) In the embodiment of FIG. 3, three example banks, **314**, **324**, and **334** are shown. Each bank is associated with a respective local address storage structure and address generator, as well as other bank specific circuitry not shown in FIG. 3 (e.g., a row decoder, RHR state control circuit). Thus, for example, the first bank **314** has a bank specific local address storage **310** and address generator **312**, the second bank **324** has a bank specific local address storage **320** and address generator **322**, and the third bank **334** has a bank specific local address storage **330** and address generator **332**. Since the bank specific circuits may generally be similar to each other, only the first bank **314** and its circuits will be described in detail.

(52) The match address HitXADD may be stored in one of the local address storage structures **310**, **320**, or **330** based on the match bank address HitBADD. For example, a bank decoder (not shown) may activate one of the address storage structures, and then the match row address HitXADD may

be stored in the activated address storage structure. The addresses HitXADD and HitBADD may, in some embodiments, be provided along a dedicated bus. The dedicated bus may operate in a serial fashion, a parallel fashion, or combinations thereof. In some embodiments, the addresses HitXADD and HitBADD may be provided along the same address bus which carries row and bank addresses (e.g., XADD and BADD) as part of normal access operations. In such embodiments, the memory may include logic which manages the timing of when the addresses HitXADD and HitBADD are provided so as not to interfere with normal memory operations.

(53) The local address storage **310** may store one or more addresses HitXADD which were associated with the value of HitBADD associated with the bank **314**. In some embodiments, the local address storage **310** may store only a single address HitXADD. In some embodiments, the local address storage **310** may store multiple addresses HitXADD. In embodiments where the local address storage **310** stores multiple addresses, logic (e.g., FIFO) may be used to manage the queue.

(54) When the bank logic (e.g., the RHR state control **236** of FIG. 2) determines that a targeted refresh operation should be performed, the address HitXADD stored in the local address storage **310** may be provided to the address generator **312**. The address generator **312** may calculate one or more refresh addresses RXADD based on the provided HitXADD. For example, the refresh addresses RXADD may represent word lines which are physically adjacent to the word lines associated with HitXADD. Other relationships (e.g., ± 2 , ± 3 , etc.) may be used. A row decoder associated with the bank **314** may then refresh the word lines associated with the refresh address RXADD.

(55) In some embodiments, the local storage **310** may be omitted, and the aggressor detector circuit **302** may provide the addresses HitXADD and HitBADD directly to the address generator.

(56) In some embodiments, different components of the memory **300** may be located in different regions of the memory chip. As indicated by dotted line **309**, some components may be located in a 'bank region' or bank logic section which is physically proximal to the associated bank. For example, the local storage **310** and address generator **312** may be located in a bank region which is physically close to the first bank **314**, the local storage **320** and address generator **322** may be located physically close to the second bank **324**, etc. In contrast, some components which are not bank specific, such as the aggressor detector **302**, may be located in a central region or central logic region of the memory chip. For example, the aggressor detector **302** may be located relatively far away from any of the banks. In some embodiments, the aggressor detector **302** may be located near the command/address pads of the memory (e.g., near the C/A terminals of FIG. 1). Putting the aggressor detector **302** in a central region may be useful as the central region may be less crowded than the bank logic regions and the aggressor storage structure **304** may take up a relatively large amount of space.

(57) In some embodiments, the use of a shared aggressor detector circuit **302** may reduce the total size of the address storage structure **304** compared to memory devices where there is not a shared aggressor detector circuit **302** (e.g., and each bank has its own aggressor detector circuit **302**). For example, the shared address storage structure **304** may store N addresses, but a memory device with no shared storage may have storage structures which store A addresses in each of B banks, and the total number of stored addresses $A*B$ may be greater than N (although in some embodiments N may be greater than A). This may be because the total number of stored addresses A may be based on a 'worst case scenario' for the bank, whereas the number N in a shared embodiment may be based on a worst case which accounts for the fact that all banks cannot have a worst case attack at the same time (e.g., due to the limitations of how accesses work in the memory).

(58) Accordingly in a shared embodiment such as the one shown in FIG. 3, the 'worst case scenario' may be based on a maximum rate at which all of the banks (e.g., **314**, **324**, **334** etc.) of the memory can be attacked, rather than being based on a rate at which any one bank may be attacked. For example, the memory may have a maximum rate at which it can be accessed. Thus, if a single bank is being attacked at a relatively high rate that may preclude accesses to other banks of the

memory. Hence, since slots of the address storage structure **304** can be dynamically allocated to different banks, the overall number of slots may be based on this maximum attack rate, since a maximum attack rate for a single bank may prevent a maximum attack rate from occurring in additional banks. Accordingly, in a shared embodiment the total number of address storage space may be less than the total number of address storage space in memory devices which do not use shared aggressor detection. This may reduce an amount of space used for address storage on the device.

(59) FIG. **4** is a block diagram of an aggressor detector according to some embodiments of the present disclosure. The aggressor detector **400** of FIG. **4** may, in some embodiments, be included in the aggressor detector **232** of FIG. **2**. Since the aggressor detector **400** may act in a manner generally analogous to the aggressor detector **232** of FIG. **2** and/or **302** of FIG. **3**, for the sake of brevity features, operations, and components previously described with respect to those figures will not be described in detail again.

(60) The aggressor detector **400** uses a hash circuit **410** to operate the aggressor storage structure **404**. In the embodiment of FIG. **4**, rather than directly storing the addresses XADD and BADD in the storage structure **404**, the aggressor storage structure may use a hash generator **410** to generate a hash value, which may be used to index a count value in the storage structure **404**. In this manner, a large number of possible values of received row and bank address XADD and BADD may be tracked by a smaller number of count values.

(61) The hash generator **410** may receive a input value based on the row and bank address XADD and BADD, and may provide an index value Hash. The input value may include a first number of bits, and the index value Hash may include a second number of bits which is smaller than the first number. Accordingly, multiple values of Input may be associated with a value of the index value Hash. Each value of the index value Hash may be associated with a count value in the storage structure **404**.

(62) Based on the value of Hash, one of the counts in the storage structure **404** may be changed (e.g., incremented). The storage logic **406** may use the count values to determine if the received row and bank address XADD and BADD should be provided as the match address HitXADD and HitBADD. For example, the storage logic **406** may compare the changed count value to a threshold, and if the count value is greater than the threshold, the received row and bank address may be provided as the match address. The count value may then be changed (e.g., reset, decreased, etc.).

(63) In some embodiments, the input value Input may include both the row and bank address XADD and BADD. For example, if the row address is 17 bits and the bank address is 5 bits, then the value Input may be 22 bits and may be concatenation of the row and bank address. In sonic embodiments, the input value Input may be the row address, and a second hash generator (not shown) may hash the bank address.

(64) FIG. **5** is a block diagram of a method according to some embodiments of the present disclosure. The method **500** may, in some embodiments, be implemented by one or more of the components, apparatuses and/or systems described herein.

(65) The method **500** may generally begin with block **510**, which describes performing an access operation by providing a row address and bank address. The row and bank address may be provided along an address bus by an address decoder (e.g., **104** of FIG. **1**). The row and bank address may be multi-bit signals, the value of which specifies a row and bank, respectively. For example, the bank address may specify one of a plurality of banks, while the row address may specify one of a plurality of rows (word lines) within that bank).

(66) Block **510** may generally be followed by block **520**, which describes receiving the row and bank address at an aggressor detector circuit. In some embodiments the row and bank address may be sampled, and may be received responsive to an activation of a sampling signal. The activations of the sampling signal may be performed with random timing, regular timing, semi-random timing,

pseudo-random timing, timing based on one or more other signals, or combinations thereof. In some embodiments, the aggressor detector circuit may be located in a central region of a memory device (e.g., near the C/A terminals).

(67) Block **520** may generally be followed by block **530**, which describes determining if the received row and bank address are an aggressor row and bank address based, in part, on a match between the received row and bank address and one of a plurality of stored row and bank addresses in the aggressor detector circuit. A storage logic circuit may compare the received row and bank address to one or more stored row and bank addresses in an aggressor address storage structure. A match may be determined if the received row and bank address match the value of a stored row address and its associated stored bank address respectively. If there is not a match, in some embodiments, the storage logic circuit may store the received row and bank address in the aggressor address storage structure. In some embodiments, if there is a match, the received row and bank address may be determined to be an aggressor row and bank address. In some embodiments, the aggressor address storage structure may include count values associated with each stored row and bank address, and the count value may be changed (e.g., incremented) responsive to a match. The received row and bank address may be determined to be an aggressor row and bank address based on the count value (e.g., a comparison of the count value to a threshold).

(68) Block **530** may generally be followed by block **540**, which describes generating a refresh address based on the received row address and providing the refresh address to a bank based on the received bank address if the received row and bank address are an aggressor row and bank address. For example, the aggressor row and bank address may be provided to a selected set of bank specific circuits based on the aggressor bank address. The bank specific circuits may, in some embodiments, include a local storage structure which may hold the aggressor row address. The bank specific circuits may include a refresh address generator, which may generate the refresh address based on the aggressor row address as part of a targeted refresh operation. The refresh address may represent word lines which have a physical relationship (e.g., adjacency) to the word line represented by the aggressor row address. As part of the targeted refresh operation, the word line(s) associated with the refresh address may be refreshed.

(69) As used herein, an activation of a sample may refer to any portion of a signals waveform that a circuit responds to. For example, if a circuit responds to a rising edge, then a signal switching from a low level to a high level may be an activation. One example type of activation is a pulse, where a signal switches from a low level to a high level for a period of time, and then back to the low level. This may trigger circuits which respond to rising edges, falling edges, and/or signals being at a high logical level. One of skill in the art should understand that although embodiments may be described with respect to a particular type of activation used by a particular circuit (e.g., active high), other embodiments may use other types of activation (e.g., active low).

(70) Of course, it is to be appreciated that any one of the examples, embodiments or processes described herein may be combined with one or more other examples, embodiments and/or processes or be separated and/or performed amongst separate devices or device portions in accordance with the present systems, devices and methods.

(71) Finally, the above-discussion is intended to be merely illustrative of the present system and should not be construed as limiting the appended claims to any particular embodiment or group of embodiments. Thus, while the present system has been described in particular detail with reference to exemplary embodiments, it should also be appreciated that numerous modifications and alternative embodiments may be devised by those having ordinary skill in the art without departing from the broader and intended spirit and scope of the present system as set forth in the claims that follow. Accordingly, the specification and drawings are to be regarded in an illustrative manner and are not intended to limit the scope of the appended claims.

Claims

1. An apparatus comprising: a plurality of memory banks, each comprising a plurality of word lines; and an aggressor detector circuit configured to receive a bank address associated with one of the plurality of memory banks and a row address associated with one of the plurality of word lines in that bank and determine if the one of the plurality of word lines is an aggressor based on the row address and the bank address, wherein the aggressor detector circuit is configured to determine if the row address and the bank address are associated with the aggressor word line based on a comparison of the row address and the bank address to a stored row address and a stored bank address.
2. The apparatus of claim 1, wherein the aggressor detector circuit includes a storage structure configured to store a plurality of count values, wherein responsive to the row address and the bank address, a selected one of the plurality of count values is updated, and wherein the aggressor detector circuit is configured to determine if the one of the plurality of word line is the aggressor based on the selected one of the plurality of count values.
3. An apparatus comprising: a plurality of memory banks, each comprising a plurality of word lines; and an aggressor detector circuit configured to receive a bank address associated with one of the plurality of memory banks and a row address associated with one of the plurality of word lines in that bank and determine if the one of the plurality of word lines is an aggressor based on the row address and the bank address, wherein the aggressor detector circuit includes a storage structure configured to store a plurality of count values, wherein responsive to the row address and the bank address, a selected one of the plurality of count values is updated, and wherein the aggressor detector circuit is configured to determine if the one of the plurality of word line is the aggressor based on the selected one of the plurality of count values, and wherein the storage structure comprises a plurality of slots each associated with one of the plurality of count values, the plurality of slots configured to store a stored row address and a stored bank address, and wherein the selected one of the plurality of count values is selected based on a match between the row address and the bank address the stored row address and the stored bank address the associated one of the plurality of slots.
4. The apparatus of claim 1, further comprising a plurality of address generator circuits each associated with one of the plurality of memory banks, wherein the row address is provided to a selected one of the plurality of address generator circuits associated with the memory banks which includes the aggressor word line based on the row address.
5. The apparatus of claim 4, further comprising: a first address bus configured to provide the row address to the one of the plurality of memory banks as part of an access operation; and a second address bus configured to provide the row address to the selected one of the plurality of address generator circuits from the aggressor detector circuit.
6. The apparatus of claim 4, wherein the selected one of the plurality of address generators is configured to generate a refresh address based on the row address.
7. An apparatus comprising: a first address generator associated with a first memory bank; a second address generator associated with a second memory bank; and an aggressor detector configured to receive a row address and a bank address, determine if the row and the bank address are associated with an aggressor word line, and provide the row address to one of the first address generator or the second address generator based on the bank address if the row address and the bank address are associated with the aggressor word line, wherein the aggressor detector is configured to determine if the row address and the bank address are associated with the aggressor address based, in part, on a comparison of the row address and the bank address to a stored row address and a stored bank address.
8. The apparatus of claim 7, wherein the aggressor detector is configured to update a count value

based on the row address and the bank address and determine if the row address and the bank address are associated with the aggressor address based on the count value.

9. The apparatus of claim 7, further comprising: a first address bus configured to provide the row address to the first memory bank or the second memory bank as part of an access operation; and a second address bus configured to couple the aggressor detector to the first address generator and the second address generator.

10. An apparatus comprising: a first address generator associated with a first memory bank; a second address generator associated with a second memory bank; an aggressor detector configured to receive a row address and a bank address, determine if the row and the bank address are associated with an aggressor word line, and provide the row address to one of the first address generator or the second address generator based on the bank address if the row address and the bank address are associated with the aggressor word line; a first local storage circuit associated with the first address generator; and a second local storage circuit associated with the second address generator, wherein the row address is stored in the first local storage circuit or the second local storage circuit based on the bank address if the row and the bank address are associated with the aggressor word line.

11. The apparatus of claim 10, wherein the first address generator configured to generate a first refresh address based on an address stored first local storage circuit and second address generator configured to generate a second refresh address based on an address stored in the second storage local circuit.

12. An apparatus comprising: a first address generator associated with a first memory bank; a second address generator associated with a second memory bank; an aggressor detector configured to receive a row address and a bank address, determine if the row and the bank address are associated with an aggressor word line, and provide the row address to one of the first address generator or the second address generator based on the bank address if the row address and the bank address are associated with the aggressor word line; and command/address terminals configured to receive the row address and the bank address, wherein the first address generator is in an area closer to the first memory bank than to the command/address terminals, the second address generator is in an area closer to the second memory bank than to the command/address terminals, and the aggressor detector is in an area closer to the command/address terminals than to the first memory bank or the second memory bank.

13. A method comprising: performing an access operation based on a row address and a bank address; comparing the row address and the bank address to a plurality of stored row addresses and associated bank addresses; determining that the row address and the bank address are an aggressor row address and an aggressor bank address based on the comparing; and storing the row address and the bank address if they do not match any of the plurality of stored row addresses and the plurality of stored bank addresses.

14. The method of claim 13, further comprising: providing the aggressor row address to an address generator associated with a bank based on the aggressor bank address; and generating a refresh address from the aggressor row address with the address generator.

15. The method of claim 14, further comprising refreshing at least one word line in the bank based on the refresh address.

16. The method of claim 14, further comprising: storing the aggressor row address in a local storage circuit associated with the bank; and generating the refresh address based on the aggressor row address in the local storage.

17. The method of claim 13, further comprising: providing the row address and the bank address to a bank along a first bus as part of the access operation; and providing the aggressor row address an address generator associated with the bank along a second bus different than the first bus.
